Experiment 8 — Alumnex Connect Business Use-Case Model

Description

Experiment 8 — Alumnex Connect Business Use Cases and Use-Case Model

Purpose and system boundary

This use-case model describes the existing Alumnex Connect platform. The source repository has a React single-page frontend, Express REST API, Mongoose/MongoDB domain persistence, Socket.IO real-time messaging, external developer/career API integrations, Google Calendar routes and a Capacitor mobile client. This is not the earlier generic student course/examination portal.

Repository: FSD/alumni-connect-website-main at master · SrimanKumarV/FSD
Live deployment entry point: AlumniConnect - Student-Alumni Platform
Route wiring evidence: backend/server.js, frontend route map.

Actors

Details

Assignee
subhadharanis.24cse
Assign to me

Parent
ACS-28 Alumnex Connect — Scrum Lab Ex

Sprint
Add sprint

Labels
experiment-8 scrum-lab

Due date
Add due date

Team
Add team

- **Student/member** — register, verify email, sign in, update profile, follow/connect with members, browse mentors, request mentorship, apply for jobs, use AI career tools, view DevPulse/activity, message members, browse events/forums/projects/contests, and manage notification/calendar preferences.
- **Alumni/mentor** — maintain professional profile and availability, receive/respond to mentorship requests, manage sessions/milestones/notes, review mentees, and post/manage suitable job opportunities.
- **Administrator** — approve/verify users and mentors, moderate content, manage roles/suspensions, send system notifications, inspect analytics and system incidents.
- **College/institution / business actor** — participate in the role-specific directory and institutional/business capabilities where configured in the app.
- **External identity providers** — Google OAuth and GitHub OAuth providers used by the auth integration.
- **Email / OTP delivery provider** — supports account-verification, password-recovery and optional 2FA communication.
- **Google Calendar** — external calendar account/resource actor for connecting, listing calendars, reading/writing events, syncing and webhook callbacks.
- **Developer platforms** — GitHub, LeetCode, Codeforces and other supported developer-data integrations.
- **AI providers** — Groq and Google Gemini provider paths referenced by the project README/source.
- **Web Push / Firebase Cloud Messaging** — browser and Android push-delivery boundaries.

The database is a persistence component, not a human actor. Actors/providers are included only where source modules/configuration define an integration; live

Start date

Add date

Story point estimate

Add story points

Reporter

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Development

Connect development tools

Open in coding tool

Create branch

Create commit

Automation

Recent rule runs Refresh

Refresh to see recent runs.

Create new automation rule

Created 1 day ago

Updated 30 minutes ago

Configure

credentials and end-to-end connectivity still require runtime validation.

Use-case inventory by business capability

1. **Identity/access:** register, log in, OAuth callback, choose role, verify email, change/reset password, optionally verify 2FA, refresh session, log out.
2. **Member network:** view/search member profiles, update own profile, connect/follow, manage connection requests and presence.
3. **Mentorship:** discover mentor, request mentorship, automatically match, respond/change status, schedule session, manage milestones/notes, cancel, review and update mentor reward.
4. **Career:** browse internal/external jobs, create/update openings when authorized, parse resume, save opportunity, apply, review application status, request AI chat/resume analysis/job match/mock interview.
5. **Developer momentum:** connect developer profile, refresh platform information, sync activity, manage goals/preferences, view timeline, streaks, analytics, insights and export/delete own activity.
6. **Communication and alerts:** start conversation, send/read messages, manage groups, search conversation, receive/mark/read notifications, subscribe/unsubscribe web push, register/unregister device.
7. **Events and calendar:** discover/register/unregister for events, manage event state by role, connect/disconnect Google Calendar, manage settings, list calendars, read/write events, sync and provide schedule context.
8. **Community and portfolio:** forum post/feed/search/comment/like/report, Tech Hub question/reply, create/manage project showcase, browse/participate in contests and leaderboards, search institution/business directory.
9. **Governance:** approve users/mentors, manage roles/status, moderate content, access admin analytics, inspect system health/incidents and send admin

notifications.

Important system relationships

- Sign-in and restricted operations require authentication/role checks in `backend/middleware/auth.js`.
- Mentorship requests interact with the mentorship domain and can produce notifications/session/review records.
- Job application relates a user to a job and application status.
- Activity sync uses developer/profile integrations and writes activity/goal records.
- Messages and notifications are persisted; Socket.IO transports real-time updates; Redis adapter enables cross-instance pub/sub.
- Calendar workflows pass through Google OAuth and per-user calendar connection/event state.
- AI endpoints call external model providers; credentials/quota availability are deployment configuration, not implied by the diagram.

Mermaid source — business use-case overview

```
flowchart LR
  Student[Student / Member]
  Alumni[Alumni / Mentor]
  Admin[Administrator]
  Org[Institution / Business]
  OAuth[Google / GitHub OAuth]
  Email[Email / OTP Provider]
  DevAPIs[Developer APIs]
  AI[Groq / Gemini AI]
  GCal[Google Calendar]
  Push[Web Push / FCM]

```

```
  subgraph AC[Alumnex Connect]
    Identity((Identity and Profile))
    Network((Member Network))
    Mentorship((Mentorship Lifecycle))
    Career((Jobs and Applications))
    AItools((AI Career Assistance))
    Activity((DevPulse and Activity Goals))
    Chat((Messages and Notifications))
    Events((Events and Calendar UI))
    Community((Forum, Tech Hub, Projects, Contests))
    AdminUse((Governance and Analytics))
  end

  Student --> Identity
  Student --> Network
  Student --> Mentorship
  Student --> Career
  Student --> AItools
  Student --> Activity
  Student --> Chat
  Student --> Events
  Student --> Community
  Alumni --> Identity
  Alumni --> Mentorship
  Alumni --> Career
  Alumni --> Chat
  Alumni --> Community
  Admin --> AdminUse
  Admin --> Identity
  Admin --> Community
  Org --> Network
  Identity --- OAuth
  Identity --- Email
  Activity --- DevAPIs
  AItools --- AI
  Events --- GCal
  Chat --- Push
```

Evidence status

Actors, business flows, system boundary and source diagram are recorded in this issue. This is a source-derived design model; it is not a claim that each external provider was successfully exercised from the live demo in this session.

Rendered diagram evidence (updated for this repository study)

The current Alumnex-specific diagram files are in `ACS_Scrum_Experiments_Artifacts.zip`:

- `diagrams/use_case_model.png`
- `diagrams/use_case_model.svg`
- `diagrams/use_case_model.mmd`
- `diagrams/use_case_model.dot`

The image files are **not attached to this Jira issue yet** because the Atlassian Media upload failed from the execution environment. Manual attachment follow-up: ACS-46. This issue remains the written use-case model source; attach the image before claiming in-Jira visual evidence is complete.

Attachments

Add attachment | Create project plan Try

Subtasks

Add subtask

Linked work items

Add linked work item

Activity

All | Comments | History | Work log

SV

Add a comment...

Suggest a reply... | Status update... | Thanks...

Pro tip: press M to comment

SV **SRIMAN VEL**
1 day ago

Artifact update: rendered use-case diagram (SVG and PNG) plus Graphviz source are included in the refreshed downloadable ACS experiment package. The actor/use-case/system-boundary model is also written directly in this issue; no live production integrations are claimed.

SV **SRIMAN VEL**
1 day ago

Use-case model is recorded in the issue description, with actors, system boundary and core use-case links. This keeps the experiment evidence